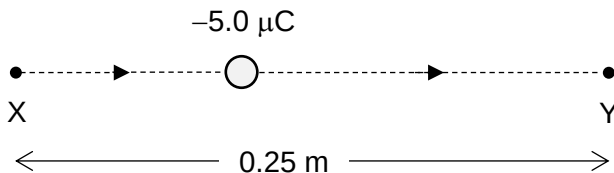


- 21 A particle of charge $-5.0 \mu\text{C}$ is projected from X towards Y with kinetic energy $250 \mu\text{J}$. When the particle is at Y which is 0.25 m away from X, its kinetic energy decreased to $150 \mu\text{J}$.



Which of the following statements may not be correct?

- A X is at a higher electric potential than Y.
- B The potential difference between X and Y is 20 V .
- C The electric field between X and Y is of uniform strength 80 N m^{-1} .
- D The electric field between X and Y is directed towards Y.

Ans: C

The field between X and Y need not be uniform. (C)

Since particle lost KE, it gained EPE + particle is negatively charged

→ X is at a higher potential than Y (A)

$\Delta V = \Delta W / q = (250 - 150) / 5.0 = 20 \text{ V}$ (B)

→ Field is directed from X to Y (D)

- 22 The diagram shows an electrical circuit consisting of a 1.5 V cell of negligible internal resistance